

Project Proposal: Streamflow Prediction at Unmonitored Rivers

Does Grouping Similar Watersheds Help Predict River Flow Where There Are No Gauges?

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Heilmeier's Criteria

What are we trying to do? Estimate how much water flows in a river each day in areas with no measuring equipment, using only characteristics of the region/water basin and local weather.

How is it done today, and what are the limits? Hydrologists tune a physical model for each river using years of measured flow. For unmeasured rivers, they borrow information from nearby or similar "donor" rivers, which often fails across climates. Deep learning models (LSTMs) trained on hundreds of rivers now do better: median NSE 0.69 on 531 unmeasured US basins ([Kratzert et al., 2019](#)), but they're costly to train and difficult to understand.

What is new, and why will it succeed? We'll test whether the traditional "similar rivers behave alike" idea improves simple tabular unsupervised ML models. We cluster basins on land, soil and climate attributes, then compare results from the following scenarios: telling a model nothing, the basin's cluster, or its full attributes. To verify our results we'll evaluate a synthetic benchmark that should model nicely with clustering, so the actual results are interpretable however they land. We'll also evaluate whether different regional climates are an important feature in informing the models we deploy.

Who cares? Most of the world's rivers have no gauge, yet water managers need flow estimates for flood and drought planning. A cheap, interpretable method, or clear evidence of when it fails, is useful either way. Application of this research is perhaps more useful in areas prone to drought, so we'll evaluate our model in regions beyond the US and the UK.

Risks and payoffs. Clusters may discard information that raw attributes keep and fixed weather windows may miss slow snowmelt and groundwater. We'll evaluate additional potential influences beyond the data present in the Caravan dataset. Climate differences may break US-to-Great Britain transfer, so it may make sense to build separate models depending on the class of region. Each risk to the analysis' evaluation and results is an explicit comparison in our design, not an assumption. The return is evidence-backed best practices for predicting flow in unmeasured basins with simpler, more interpretable models.

Methods, Data and Evaluation

Methods. *Baselines:* Ridge regression on lagged weather measurements and accuracy of nearest-gauge transfer. *Unsupervised:* PCA, then k-means vs. agglomerative clustering with varying k. *Supervised:* LightGBM and RBF kernel regression (Nyström approximation; exact SVR on a subsample). *Basin information given to the model:* none; cluster ID; one model per cluster; PCA or raw attributes. *Preprocessing compared:* weather windows (7/30 vs. adding 90/365 days), raw vs. log flow, PCA on vs. off. *Hyperparameters:* k, tree depth and learning rate, kernel bandwidth.

Data. Real data include two Caravan subsets in the same format: CAMELS-US (150–300 basins) and CAMELS-GB plus a third, drought-prone region, with daily weather, daily flow and static basin attributes. We'll exclude any attribute derived from observed or modeled flow (e.g., mean flow, runoff ratio), which could leak the answer. We'll also synthesize data with parameters sampled from real data to evaluate the model on specific scenarios.

Success. Answering the central question across basins: does cluster information improve held-out median NSE over full attributes? Secondly: can we achieve held-out median NSE above 0.40, and a positive absolute NSE gain over nearest-gauge transfer.

Tools. Python (pandas, scikit-learn, LightGBM, SHAP, matplotlib)